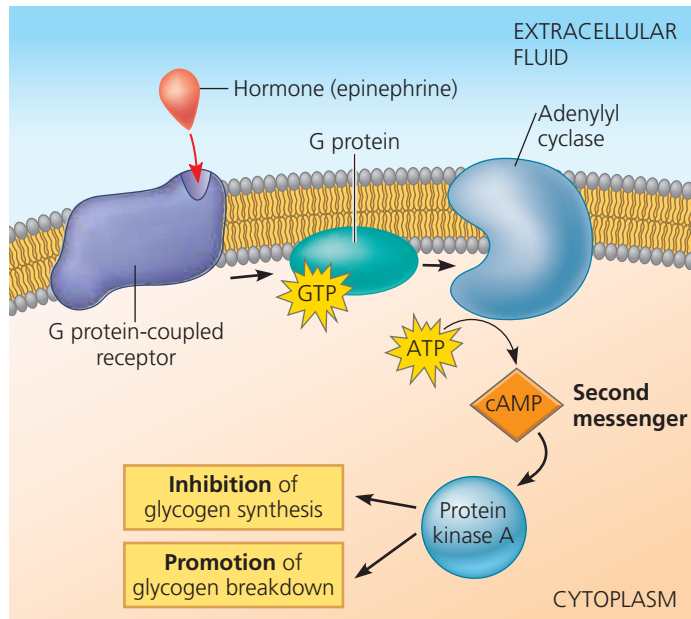


▼ **Figure 45.6** Signal transduction triggered by a cell-surface hormone receptor.



VISUAL SKILLS A series of arrows represents the steps linking epinephrine to protein kinase A. How does the event represented by the arrow between ATP and cAMP differ from the other four?

➔ **Mastering Biology Animation: Water-Soluble Hormone Pathway**

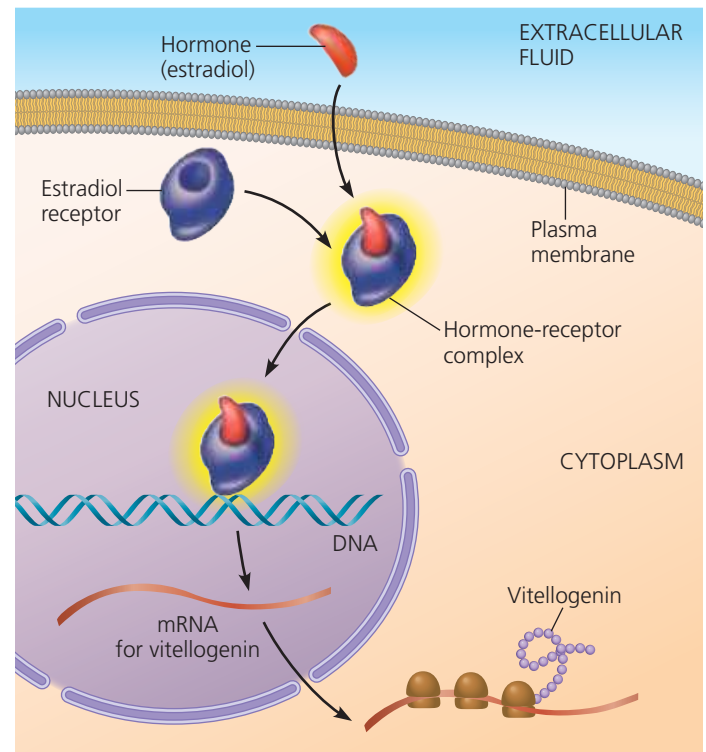
Response Pathway for Lipid-Soluble Hormones

Intracellular receptors for lipid-soluble hormones perform the entire task of transducing a signal within a target cell. The hormone activates the receptor, which then directly triggers the cell's response. In most cases, the response to a lipid-soluble hormone is a change in gene expression.

Most steroid hormone receptors are located in the cytosol prior to binding to a hormone. Binding of a steroid hormone to its cytosolic receptor forms a complex that moves into the nucleus (see Figure 11.9). There, the receptor portion of the complex interacts with a specific DNA-binding protein or response element in the DNA, altering transcription of particular genes. (In some cell types, steroid hormones trigger additional responses by interacting with other kinds of receptor proteins located at the cell surface).

Among the best-characterized steroid hormone receptors are those that bind to estrogens, steroid hormones necessary for female reproductive function in vertebrates. For example, in female birds and frogs, estradiol, a form of estrogen, binds to a cytoplasmic receptor in liver cells. Binding of estradiol to this receptor activates transcription of the vitellogenin gene (**Figure 45.7**). Following translation of the messenger RNA, vitellogenin protein is secreted and transported in the blood to the reproductive system, where it is used to produce egg yolk.

▼ **Figure 45.7** Direct regulation of gene expression by a steroid hormone receptor.



➔ **Mastering Biology Animation: Steroid Hormone Pathway**

Thyroxine, vitamin D, and other lipid-soluble hormones that are not steroids typically have receptors in the nucleus. These receptors bind to hormone molecules that diffuse from the bloodstream across both the plasma membrane and nuclear envelope. Once bound to a hormone, the receptor binds to specific sites in the cell's DNA and stimulates the transcription of specific genes.

Multiple Responses to a Single Hormone

Although hormones bind to specific receptors, a particular hormone can vary in its effects. A hormone can elicit distinct responses in particular target cells if those cells differ in receptor type or in the molecules that produce the response. In this way a single hormone can trigger a range of activities that together bring about a coordinated response to a stimulus. For example, the multiple effects of epinephrine form the basis for the "fight-or-flight" response, a rapid response to stress that you'll read about in Concept 45.3.

Endocrine Tissues and Organs

Some endocrine cells are found in organs that are part of other organ systems. For example, the stomach contains isolated endocrine cells that help regulate digestive processes by secreting the hormone gastrin. More often, endocrine cells are grouped in ductless organs called **endocrine glands**,